Divya Murugan

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I am an enthusiastic engineer, always keen to learn new skills and exhibit them in challenging environment to deliver great solutions.

Overall Experience: 7 years 1 month

Team Leader: 5 years

Professional Experience:

Hitachi Energy Technology Services Private Limited **Chennai**
R&D Engineer **March 2022-December 2025**

- Working for FPGA BFM(Bus Functional Model) Design and Development, Testbench Development and RTL Design and Development in **MACH(Modular Advanced Control for HVDC and FACTS).**
  HVDC - High Voltage Direct Current
  FACTS - Flexible Alternating Current Transmission System
- Hands-on experience in tools like **ModelSim, Xilinx Vivado.**
- Basic Knowledge in tool like **HiDraw, Python.**
- Hands-on experience in programming like **VHDL.**
- Basic knowledge in programming like **Python Scripts.**
- Hands-on experience in **Ethernet, DMA, MII RMII Routing, PCIe, SFP, Flash, Ethercat, PTP, EEPROM, Watchdog and Interrupts.**

Indian Institute of Space Science and Technology **Thiruvananthapuram**
Junior Research Fellow **May 2019– February 2022**

- Worked in **Design and Development** of a **NAVIC receiver (IRNSS programme).**
  NAVIC - Navigation with Indian Constellation
  IRNSS - Indian Regional Navigation Satellite System
- Worked in **RTL Design and Development** of a **NAVIC receiver (IRNSS programme).**
- Developed the Correlator of a NAVIC receiver and owned ownership for them.
- Hands-on experience in tools like **Xilinx Vivado, SDK, Cadence encounter RTL Compiler, MATLAB.**
- Hands-on experience in **RTL Design and Development, Synthesis, Implementation and bitstream generation**. Finally **Programmed the board** and **tested with real time environment**.
- Hands-on experience in **Clock Domain Crossings** and **Timing Analysis.**

Signals and Systems India Private Limited **Chennai**
Junior Engineer Project and Development **January 2018 – July 2018**

- Worked in embedded firmware development of a function generator.
- Involved in design, review, documentation and testing of the Three phase reference standard meter.
- Hands-on experience in Microcontroller programming in **Embedded C.**
- Work experience in resolving issues observed.

Education:

Bachelor of Engineering – 8.44(CGPA) 2012- 2016
Kings Engineering College Chennai
Course: Electronics and Communication

Higher Secondary -85.17% 2010
W.P.A.Soundarapandian Higher Secondary School Chennai

SSLC -81.4% 2008
Kamaraj Matriculation School Chennai

Projects:

- **Hardware implementation of TCP/IP over VLC for Li-Fi Communications (Bachelor degree project)**

Developed data transfer between Server and Client over visible light communications.

Professional Skills and competences:

- Hands-on Proficiency in **RTL Design and Development** for complex blocks of an SOC and Subsystems Integration Using **VHDL**.
- Full understanding of **VLSI Design Methodologies** and tools including **VHDL**, **Verilog**, **System Verilog**, simulation, synthesis and formal verification.
- Knowledge of FPGA architecture and IO Constraints.
- Hands-on Proficiency in IP Design and Development.
- **Protocols**: **AXI**
- Experience in SOC Interconnect protocols like AXI.
- **Familiar** with **FPGA's – Xilinx ZYNQ SOC, SPARTAN 6** and **ARTIX 7.**
- Communication protocols including **I2C, SPI, USART, UART, MDIO**.
- Experience in **PCI Express** and **Ethernet Protocols**.
- Excellent logic design concepts.
- Excellent Verification concepts.
- Experience in control path-oriented designs like asynchronous **FIFO, DMA** architecture, **Shared Memory** interface design.
- Experience in creating code coverage and functional coverage plans and reports.
- Experience with tools like **VIVADO, SDK, ModelSim, Questasim, Cadence Encounter RTL Compiler.**
- Good hands on with **Logic Analyzer, Function Generator and Digital Signal Oscilloscopes**.
- Good hands on with real time verification of RTL Codes with Hardware.
- Excellent knowledge of Linux.
- Experience with **automation** using scripting techniques such as **MATLAB** Script And Basic knowledge in Python Script.
- Experience in automation tools like **MATLAB**, Python (Basic).
- Good C, Embedded C programming skills, analytical, debugging and Problem Solving skills.
- Experience with tools like Keil, LPCXpresso.
- **Familiar with Microcontrollers and Microprocessores like** 8051, ARM cortex M0+, LPC2378, LPC2364.
- Basic knowledge in Tools like Proteus, Scilab, and PSpice.
- Basic Programming skills in C++, Data structures, Assembly language.
- Experience in **Agile Methodology**.

Interpersonal Skills:

- Ability to rapidly build relationships and set up trust.
- Teamwork, Leadership, Interest to learn new skills.
- Ability to work independently and good interpersonal skills.
- Confident and determined.
- Ability to cope up with different situations.

Achievement:

- Published a Research paper in International Journal of Advanced Research Trends in Engineering and Technology (IJARTET).

Certificates:

- Participated Industrial Visit programme in “Digital transmission centre, Tax network monitoring centre and OFC cable splicing” at BSNL.
- Participated in “International Conference on Innovations and Challenges in Engineering and Technology ICICET’16”.
- **Completed RISC-V ISA & RV321 RTL Design (RISC -V Instruction Set Architecture, RISC -V RV321 RTL Architecture Design, RISC-V RV321 5 Stage Pipelined RTL Design)** Course in **MAVEN SILICON** Institute.
- **Completed VLSI Design Methodologies Certification Course** in **MAVEN SILICON** Institute.



- Completed Advanced ASIC Verification Certification Course (System Verilog, UVM-Universal Verification Methodology, PERL) in MAVEN SILICON Institute.



- **Completed Static Timing Analysis Certification Course in MAVEN SILICON Institute.**



Interests:

- VLSI Design and Verification Methodologies, Communication Technology, Embedded Systems, Programming, Hardware Design, Developing Algorithm, Emerging trends and technology.

Languages:

- ✓ English - Professional Working Proficiency
- ✓ Tamil - Native Proficiency
- ✓ Hindi - Elementary Proficiency

Declaration:

I hereby declare that the above mentioned information is true to the best of my knowledge and I bear the responsibility for the correctness of the mentioned particulars.

Place: Chennai. Yours faithfully,

DIVYA.M